CLIMATE-SMART HEALTH SECTOR FINANCING

The African Context – Do not cite

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Abstract

Africa's health systems, already grappling with funding constraints, infrastructure deficits, and a high disease burden, face an additional challenge of climate change. It exacerbates health risks through extreme weather and the spread of diseases, while the health sector itself is a significant, overlooked contributor to global greenhouse gas emissions. This creates a critical nexus where strategic intervention can yield dual benefits.

This paper argues that Africa can transform this challenge into a strategic opportunity by leveraging climate finance to build resilient, low-carbon health systems. It proposes a multi-pronged approach: integrating climate resilience into national health policies, greening health sector operations (from energy and waste management to supply chains), and, most critically, tapping into innovative climate funding streams.

The analysis details how instruments like carbon credits, green bonds, and adaptation grants can be strategically deployed to finance renewable energy for clinics, energy-efficient infrastructure, and climate-proofing of health facilities. By blending these climate funds with public and private capital, African nations can unlock billions in new investment. This approach not only reduces the sector's carbon footprint and operational costs but also generates new revenue streams, ultimately creating healthcare systems that are both healthier for populations and more resilient in the face of climate change. The paper concludes with concrete policy recommendations for governments and development partners to catalyze this transition, turning the imperative of greening into a powerful engine for health system strengthening.

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1 Introduction

Climate change is exacerbating health risks and further eroding the already fragile foundations of Africa's health system. In parallel, the health sector's direct (Scope 1 emissions) and indirect (Scope 2 and 3 emissions) contributions to greenhouse gas (GHG) emissions, present an opportunity to deploy climate financing and investments in support of Africa's fledgling health systems which are characterized by limited access, sub-standard quality, and inadequate financing. This paper proposes explores how African countries can attract a blend of public (e.g., Adaptation Finance) and private financing (e.g., green bonds, carbon credits) to strengthen health infrastructure and service delivery, reduce healthcare's carbon footprint and enhance resilience to climate change.

2 Africa's Health Sector Landscape

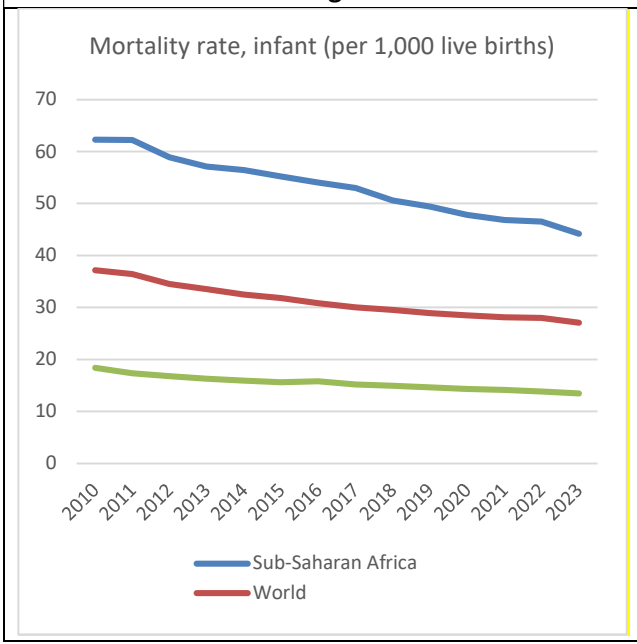
Africa has made significant strides on key health indicators but these gains risk reversal without scaled up financing to respond to the emerging challenges of climate change, population growth and the threat of future pandemics. Since 2015, life expectancy in Africa has increased by three years, maternal mortality has declined 7.4 percent, access to skilled birth attendants is up 14 percentage points, and there have been notable strides in combating infectious diseases such as HIV, tuberculosis, and malaria (UNECA Africa Sustainable Development Report 2024). Despite these gains, the continent's health system is plagued by funding constraints, limited access, weak and unreliable infrastructure, staffing shortages, a high disease burden, inefficiencies and climate change. These challenges have contributed to sub-optimal health outcomes relative to the global average.

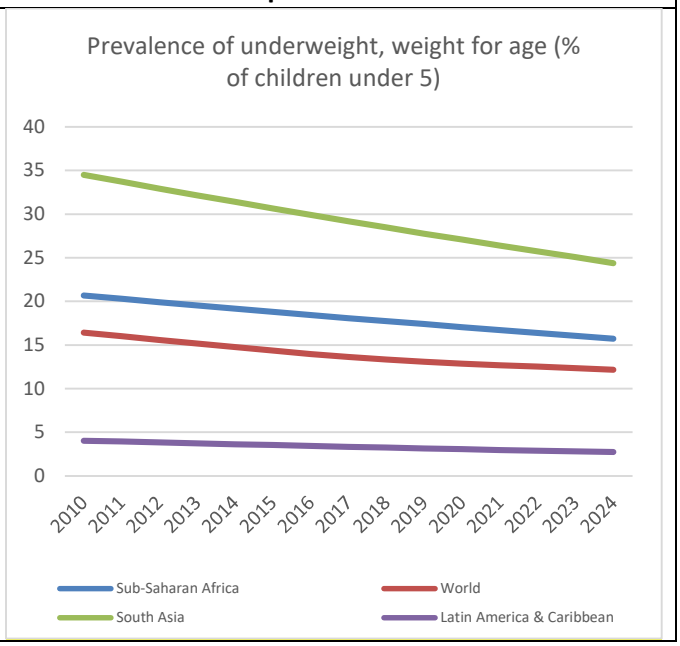
2.1.1 Key challenges

2.1.1.1 Access to quality health care:

The World Health Organization (WHO) estimates that up to 50 percent of Africans lack access to essential healthcare resulting in high infant and maternal mortality rates and other poor health outcomes (WHO, 2022). In 2023, the sub-continent suffered 44.2 deaths per 1,000 live births, almost double the global figure of 27.1 per 1,000 live births (World Development Indicators).

Figure 1: Infant mortality rates in SSA are almost double the global rate	Figure 2: Declining but relatively high underweight prevalence





Source: World Development Indicators. Last Updated 07/01/2025

2.1.1.2 Infrastructure and medical staffing shortages:

The fragility of Africa's health system is compounded by weak infrastructure and sanitation services, and inadequate skilled healthcare providers. In several African countries, access to qualified medical staff as well as critical health infrastructure such as medical supplies, energy and basic sanitation services is limited.

With respect to infrastructure, Africa's annual infrastructure needs are estimated to be between $130 and $170bn with a financing gap of $68 to $108 billion. National health systems in Africa face substantial infrastructure deficits of approximately $66 bn annually with Africa Center for Disease Control (Africa CDC) reporting an annual funding gap of USD 2 billion in 2024 (Africa CDC, 2024; UNECA, 2019).

Access to basic sanitation services is critical for the prevention and transmission of diseases. As of 2022, only 52 percent of the population of Africa had access to basic sanitation services compared to the global average of 81 percent in 2022 (UNECA Africa Sustainable Development Report 2024).

Access to reliable energy, a critical input for the effective functioning of healthcare systems, is a major challenge in Africa. As of 2023, only 53 percent of Africa's population had access to electricity (World Development Indicators WDI). A survey of 13 health facilities in 11 sub-Saharan African countries found that on average, 26% reported **no access to electricity** and only 28%, had **reliable access to electricity** among the 8 countries reporting data. These findings point to significant energy deficits in Africa's healthcare sector (Adair-Rohani H, et. al., 2013).

The lack of reliable access to energy has led to increased dependence on diesel generators which account for 25 percent of the energy mix of health facilities on the continent. Among 9 countries surveyed, approximately 7% of facilities relied solely on a generator. While this data represents only about one-quarter of sub-Saharan African nations, the sample includes 6 of the 10 most populous countries (Nigeria, Ethiopia, Tanzania, Kenya, Uganda, Ghana) (Adair-Rohani H, et. al, 2013).

2.1.1.3 Limited access to health staff

Despite bearing 25 percent of the global disease burden, access to qualified physicians and specialist surgeons is also woefully inadequate in Africa. The number of Physicians per 1000 people in Sub-Saharan Africa is 0.23 (2020) while the number of specialist surgeons per 100,000 of the population is a mere 1.7, compared to the global average of 30 per 100,000 and the Latin America and Caribbean average of 43.3 per 100,000 (WDI database).

Figure 3: SSA has fewer than 1 Physician per 1000 people

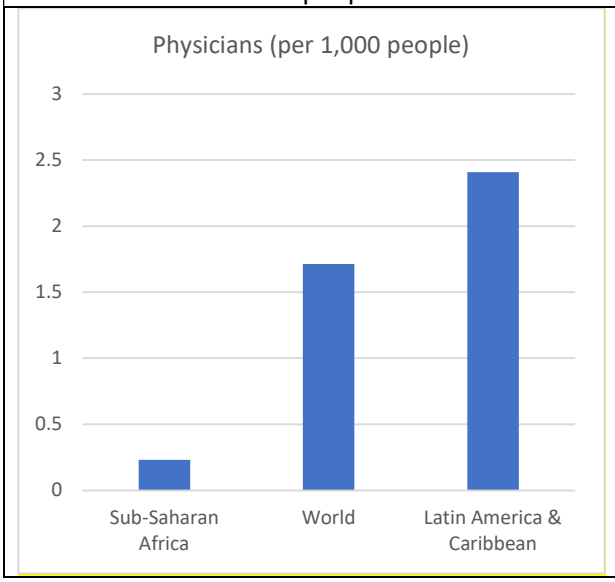
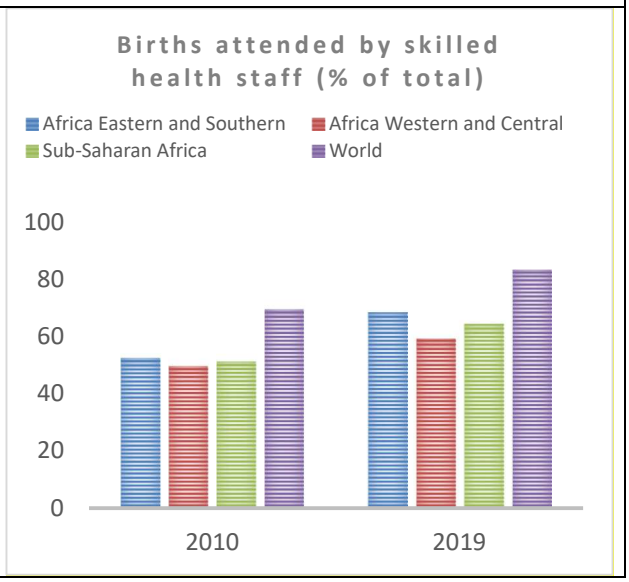


Figure 4: Eastern and Southern Africa lead in skilled birth-attendants



Source: World Development Indicators. Last Updated 07/01/2025

Furthermore, while there has been some progress in access to skilled birth attendants since 2010, as of 2019, only 64 percent of births in SSA are attended by skilled health staff compared to the global average of 84 percent (World Development Indicators).

The continent is also highly dependent on imported pharmaceuticals and medical supplies; approximately 80 to 100% of essential drugs, vaccines, diagnostics and supplies in Africa are sourced externally (Africa CDC, 2025).

2.1.1.4 Climate Change Exacerbates Health Risks

The challenges of Africa's health systems are compounded by the destructive effects of climate-induced **extreme weather events** on health infrastructure, and the rising prevalence of climate-sensitive diseases. Vector-borne diseases, such as malaria and dengue, are expanding into previously unaffected regions due to changing temperature and precipitation patterns. Similarly, respiratory illnesses, including asthma and chronic obstructive pulmonary disease (COPD) are on the rise due to worsening air quality and increased exposure to allergens like pollen. Approximately 168 per 100,000 deaths in Sub-Saharan

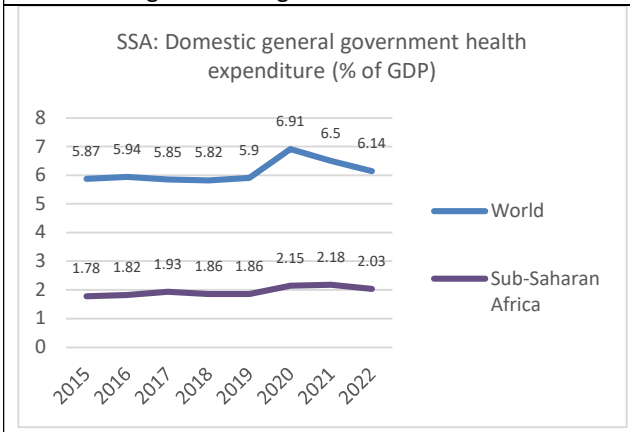
Africa are attributable to ambient air pollution, substantially higher than the world average of 104.3 and the Latin America & Caribbean region average of 41 (World Development Indicators, 2025).

Climate shocks are also fueling a rise in food insecurity and malnutrition which will in turn undermine the health of the population and further strain health services. Approximately 22 percent of the SSA population is undernourished, which is more than double the global average of 9.1 percent (WDI, 2022). Without scaled up responses, these figures could rise and further compromise the health conditions of current and future generations.

2.1.1.5 Financial Barriers and donor dependency:

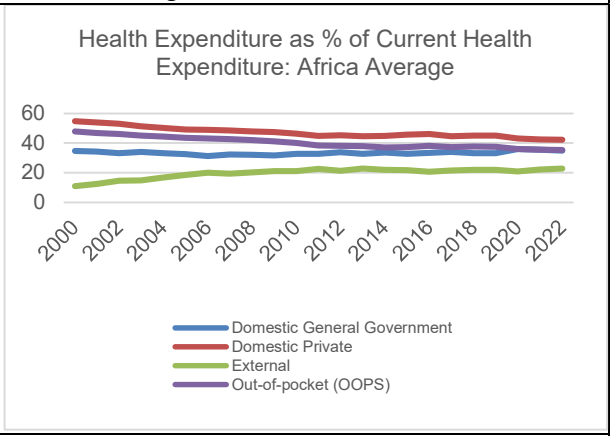
Rising debt levels, fiscal consolidation and inadequate investments in healthcare infrastructure have undermined access to quality healthcare services in Africa. As of 2022, domestic general government spending on health (i.e., excluding donor funding and private expenditure) in Sub-Saharan Africa accounted for just about 2 percent of GDP and remained largely unchanged even during the COVID-19 pandemic (Figure 5). This figure represents one third of the global average of 6.0 percent of GDP. In the same year, on a per-capita basis, Sub-Saharan governments spent USD34.17 on health financing versus the global average of USD 768.48 (WDI, 2025).

Figure 5: Domestic government health financing is a third of the global average



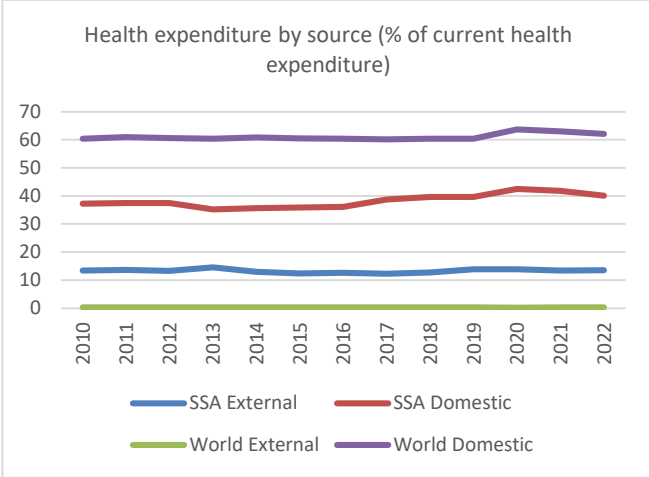
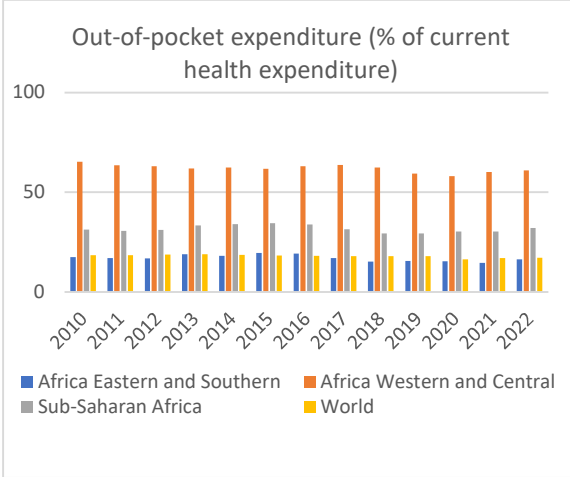
Source: Author's calculations based on World Development Indicators. Last Updated: 07/01/2025
Notes: SSA: Sub-Saharan Africa

Figure 6: Domestic private finance is Africa's dominant health financing source



Source: Author's calculations based on WHO Global Health Expenditure Database (GHED)

Domestic private insurance accounts for the largest share of health spending in Africa (42 per cent), followed by out-of-pocket spending (35 per cent), domestic government spending and external financing (22 per cent). In 2022, domestic private health insurance expenditure in sub-Saharan Africa accounted for a higher share of current health expenditure than the global average. Health sector financing from external sources was also higher than the global average of less than 1 percent (Figure 7). Overall, relative to the global average, Africa depends much more on external support (i.e., external health financing) and much less on domestic government (i.e., domestic government financing) to finance health expenditure (Figure 7).

Figure 7: Africa's health financing depends less on government and more on externals than the global average	Figure 8: Africa's high OOP is concentrated in Western and Central Africa
	

Source: Author's calculations based on World Development Indicators. Last Updated: 07/01/2025
Notes: SSA: Sub-Saharan Africa

Furthermore, relative to the rest of the world, households in SSA bear a disproportionate burden of health care costs; at least thirty percent of health costs are financed out of pocket. This figure doubles in Western and Central Africa where it accounts for at least 60 percent of health sector spending (Figure 8). A large informal sector and low levels of social protection coverage in these sub-regions may account for the relatively high out-of-pocket spending in these sub-regions.

2.1.1.6 Siloed financing – inadequate co-financing of health and climate initiatives

The strong interlinkages between health outcomes and climate change suggest the need for integrated financing of health and climate interventions to leverage the synergies between the sectors. However, national climate investments infrequently reach the health sector. For instance, half of countries surveyed by WHO reported receiving minimal or no funding from their national health budgets to implement climate and health strategies (WHO 2019).[1]

3 Africa-led initiatives to strengthen the health sector

The confluence of rising healthcare needs, inadequate financing and the compounding effects of climate change, places Africa's healthcare system at an inflection point. Several major initiatives have been launched to improve health outcomes in Africa through increased financing. The most notable is the Abuja Declaration. Launched in April 2001, by the Heads of State of African Union member countries, the initiative set a target of allocating at least 15% of national annual budgets to improve health sector outcomes. However, there has been limited progress on this front.

[1] (https://www.thinkglobalhealth.org/article/closing-gap-health-related-climate-financing)

Disaggregating the sources of health spending provides insights into the components of health spending that need to be strengthened to reduce the substantial financial burden that out-of-pocket expenditures impose on households. Since its launch in 2001, no more than 4 countries have met the Abuja Declaration target based on **domestic government financing** alone (i.e., excluding private, out-of-pocket and external financing). The peak of 4 countries (Botswana, Cabo Verde, Sao Tome and Principe and South Africa) was achieved in 2022. This figure increases to 29 countries when private insurance spending is taken into account and further rises to 45 countries when all sources of health expenditure are considered (WHO GHED).

Figure 9: Domestic government spending has had limited impact on meeting the Abuja target

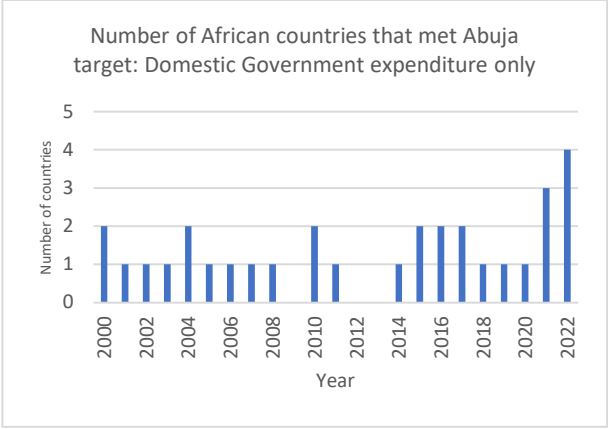
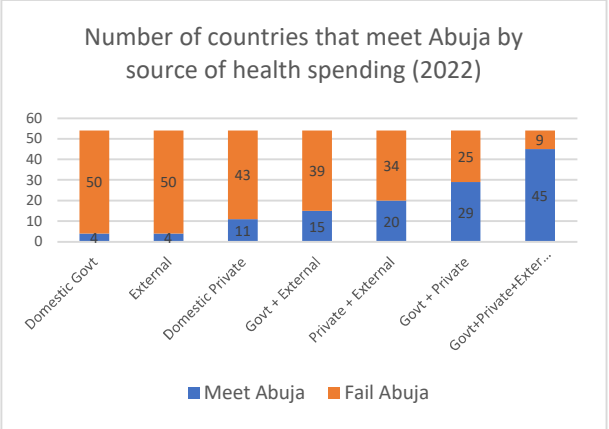


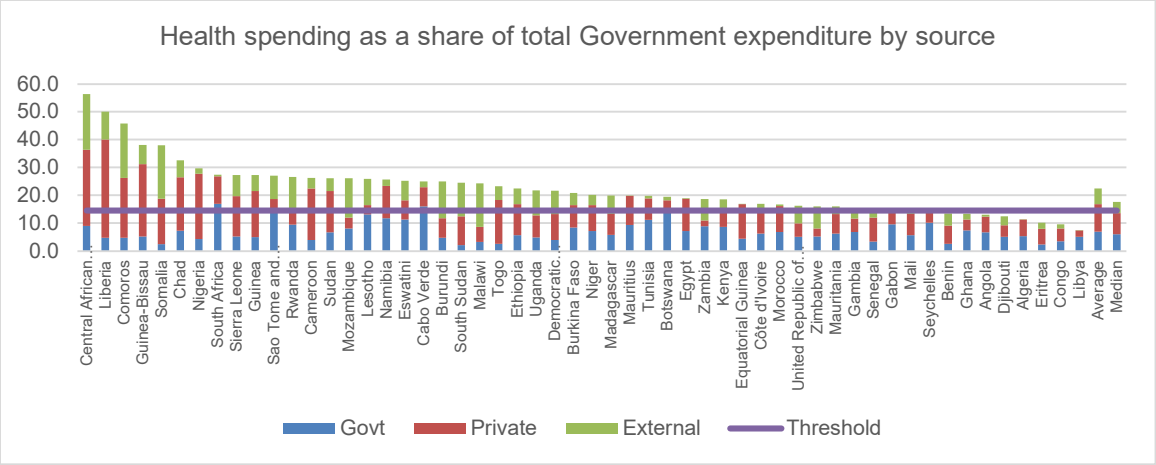
Figure 10: Thrice as many countries meet Abuja solely from private financing than government financing



Source: Author's calculations based on WHO Global Health Expenditure Database (GHED), Update December 2024

Declining ODA flows fueled by the ongoing geopolitical fragmentation and tensions will further undermine the predictability, adequacy and sustainability of Africa's external health financing. Health-related ODA to Africa in 2025 is expected to be 70 percent lower than in 2021, falling from $80 billion to a mere $24 billion (Africa CDC, 2025). Identifying measures to scale-up domestic government spending is vital to reducing dependence on external health financing.

Figure 11: 45 (83%) African countries meet Abuja based on all sources of financing (2022)



Source: Author's calculations based on WHO Global Health Expenditure Database (GHED), Update December 2024

Since the Abuja Declaration, African leaders have launched a series of initiatives to strengthen the continent's health sector. In 2019, they adopted the Africa Leadership Meeting (ALM) Declaration[2] which called on Member States to progressively increase domestic investment in health, reduce catastrophic out-of-pocket expenditures, and build resilient health systems.

Building on the ALM Declaration, AUDA-NEPAD launched the Programme for Investment and Financing in Africa's Health Sector (PIFAH) in September 2024 to mobilize and channel public, private, and blended capital, as well as institutional capacity, to support scalable health investments across Africa.

In parallel, the Africa Center for Disease Control (Africa CDC), has advanced a three-pillar strategy to strengthen health sector financing. Pillar One focuses on strengthening domestic financing to reduce external aid dependence and build self-reliance. Pillar Two introduces innovative financing mechanisms to bridge urgent gaps. Pillar Three aims to leverage blended finance to address infrastructure and supply chain gaps, catalyzing investments in areas like renewable energy for health facilities, connectivity and digital health, and local pharmaceutical manufacturing (Africa CDC, Africa's Health Financing in a New Era. Concept Paper, April 2025).

The ongoing initiatives focus largely on scaling traditional financing sources for the health sector. However, exploiting the nexus between the health sector and climate change opens up untapped climate financing opportunities for the health sector.

[2] Also known as the *Addis Ababa Commitments towards Shared Responsibility and Global Solidarity for Increased Health Financing*

4 The nexus between climate change and the health sector

Health sector outcomes and climate change impacts are enduring and mutually reinforcing. While climate change adversely burdens health systems through its adverse effects on infrastructure and the health of households, the healthcare sector is a major contributor to climate change, accounting for 4.4% of global net emissions (Health Care Without Harm, 2020). Energy consumption, transportation, as well as the manufacture, use, and disposal of medical supplies, equipment and services are the key sources of health sector emissions. If the health sector were a country, it would be the fifth-largest emitter on Earth[3].

4.1 Supply-chain emissions are the greatest source of health sector emissions

Supply chain emissions from the production, transport and disposal of goods and services such as pharmaceuticals, medical devices, food, agricultural products, medical services, and hospital equipment, constitute the largest (71 percent) source of the health care sector's footprint (Scope 3 emissions) (Health Care Without Harm, 2020).

The second largest (i.e., 17 percent) source is emissions generated **directly** from health care facilities and care-owned vehicles (Scope 1). While **indirect emissions** from purchased energy such as electricity, steam, cooling, and heating (Scope 2 emissions) represent the third largest (i.e., 12 percent) source of the sector's carbon footprint (Health Care Without Harm, 2020).

The **production and use of energy** is the largest (53 per cent) component of the health sector's carbon footprint. Most of this footprint (40%) derives from the purchase of energy by health facilities (Scope 2) as well as the energy embodied in the goods and services supplied to health care facilities (i.e., energy purchased by manufacturers and suppliers of goods and services such as pharmaceuticals and medical device industries – Scope 3 emissions). Another 13% of health care's footprint primarily arises from on-site power generation in health care facilities (Scope 1) (HCWH, 2020).

Generally, health sector emissions tend to rise with the level of health sector spending; however, this relationship depends on the energy intensity of a country's economy and the relative share of fossil fuels in the country's energy mix.

5 A Multi-Level Approach to Greening the Health Sector

The significant contribution of the health sector to global greenhouse emissions presents both a challenge and an opportunity. While the scale of Africa's health sector emissions is substantially lower than

[3] This is based on data for one year (2014). Health Care Without Harm, 2020).

developed countries, the adverse health effects pose a reputational risk to a sector that is entrusted with improving health outcomes. The opportunities arise from the financial resources that can be mobilized by greening the sector. This section proposes a three-pillar approach to greening. The first pillar is foundational and focuses on securing commitment of leadership (Foundation) and key healthcare policymakers. The second pillar is rooted in transforming the core operations of the healthcare sector. The third pillar is geared to changing the way the industry undertakes patient care (Systemic Integration) (Table 1).

Table 1: Summary of the Approach

Pillar	Focus Area	Key Entry Points
1. Foundation	Governance & Strategy	• Secure leadership buy-in and commitment, • Baseline emissions, • Green procurement policies.
2. Operational	Facilities & Operations	Invest in: • Energy efficiency & renewables, • Waste segregation & reduction, • Water conservation, • Sustainable transport.
3. Systemic	Clinical Care & Resilience	• Adopt low-carbon clinical pathways, • Engage firms in the supply chain • Build climate resilience.

5.1 The Foundation- Governance, Leadership, and Measurement

Commitment and buy-in at the highest level of leadership empower operational changes, ultimately leading to a transformative shift in how healthcare is delivered. This is achieved through leadership and commitment to change, baselining and measuring the health sector's emissions and greening procurement policies to integrate environmental criteria into tender processes and encourage manufacturers and suppliers to develop greener products.

5.1.1 Commit, Lead and Design a Health-sector Climate Strategy:

The entry point is securing high-level commitment (e.g., from the Ministry of Health, hospital CEOs, and board members) to implement a formal **Sustainability or Climate Strategy** for the health sector or individual health facility. The strategy should contain ambitious but achievable targets for decarbonization and waste reduction. For instance, The National Health Service (NHS) in England established its "Greener NHS" campaign with a legally binding target to reach **net zero by 2040**. This top-down mandate provides the authority and direction for all subsequent actions.

5.1.2 Measure and Baseline Implementation of the Strategy:

Tracking the implementation of the strategy requires conducting a comprehensive environmental footprint assessment. This involves using tools like the **NHS Carbon Footprint Calculator** or other Life Cycle Assessment (LCA) methodologies to measure the baseline data for:

- **Carbon Emissions:** Scopes 1 (direct, e.g., boilers, ambulances), 2 (indirect, from purchased electricity), and critically, **Scope 3** (supply chain, which can account for approximately 70% of a health system's footprint) emissions.
- **Resource Use:** Energy, water consumption, waste generation by type (general, hazardous, pharmaceutical).

5.1.3 Establish Green Procurement Policies:

The health sector is a massive consumer of goods and services, hence, it **can l**everage its purchasing power to steer the greening of its supply chain by integrating environmental criteria into tender processes. This encourages manufacturers and suppliers to develop greener products.

Examples include prioritizing suppliers who:

- Offer products with less packaging.
- Take back devices for refurbishment or recycling (e.g., Philips' circular model for medical equipment).
- Provide energy-efficient medical devices.
- Offer safer alternatives to harmful chemicals (e.g., mercury-free thermometers, safer disinfectants).

5.2 Core Operational Entry Points

The operational level is where tangible, high-impact changes are made within **healthcare facilities and systems**. This is because the operations of health facilities involve the production and use of energy, which is often the largest source of emissions and cost. To this end, promoting energy efficiency and renewable energy, ensuring sustainable waste management, implementing water conservation practices and achieving sustainable transportation are key to substantially reducing the carbon footprint of the healthcare sector.

5.2.1 Energy Efficiency and Renewable Energy:

Energy efficiency and renewable energy interventions involve:

- **Retrofitting existing buildings:** Switching to **LED lighting**, installing smart thermostats, and optimizing HVAC systems.
- **Building green:** Adopting green building standards (e.g., **LEED** or **BREEAM**[4]) for new constructions, ensuring natural light, efficient insulation, and energy-efficient design.
- **Generating on-site renewable energy:** Installing **solar panels** on hospital rooftops or car parks.
- **Purchasing off-site renewable energy:** Signing Power Purchase Agreements (PPAs) for wind or solar energy.

[4] LEED (Leadership in Energy and Environmental Design) and BREEAM (Building Research Establishment Environmental Assessment Method) are globally recognized, voluntary green building certification systems that evaluate and rate the sustainability of buildings and developments. LEED is a U.S.-based program popular worldwide, whereas BREEAM is the dominant system in the UK and Europe.

5.2.2 Sustainable Waste Management:

Healthcare waste is often highly polluting and expensive to dispose of. Greening waste management processes involves:

- **Segregation at source:** Properly separating general, infectious, hazardous, and pharmaceutical waste prevents over-treatment (e.g., incinerating general waste) and reduces costs.
- **Reducing and Reusing:** Replacing single-use items with reusable alternatives where safe and possible (e.g., surgical gowns, stainless steel instruments). For instance, the Aravind Eye Care System in India drastically reduces waste and cost by reusing surgical gowns, caps, and masks, which are laundered daily for reuse. This practice is part of a larger, highly efficient and environmentally sustainable model that reuses and recycles without compromising patient safety. For a single cataract surgery, the system generates only about 250 grams of waste, a fraction of the waste produced by the same procedure in the United States. (Thiel CL, et., al., 2017).
- **Recycling Programs:** Implementing robust recycling for paper, cardboard, plastics, and even metals from specific waste streams.
- **Safe Treatment:** Replacing inefficient or polluting incinerators with modern, cleaner technologies like autoclaving (steam sterilization) for infectious waste.

5.2.3 Sustainable Transportation:

Reducing the health sector's carbon footprint in transportation involves investments in:

- **Electric vehicle fleets:** Transitioning ambulances and staff vehicles to electric vehicles (EVs) reduces operational costs and the carbon footprint of the health facility.
- **EV charging stations:** installing EV charging stations for staff and patients, supports the transition to electric vehicles.
- **Public transport, cycling, and telemedicine:** minimizing the number of vehicles on the road reduces carbon emissions as well as the carbon footprint of patient and staff travel.

5.3 Systemic Integration- Embedding Sustainability into Care

Greening the processes associated with patient care is the most advanced step towards integrating environmental health with clinical health. This includes low-carbon clinical care, ensuring a sustainable supply chain, and building climate resilient healthcare systems.

5.3.1 Low-Carbon Clinical Care:

Achieving low-carbon clinical care requires engaging clinicians to identify and implement changes in clinical pathways that reduce environmental impact without compromising care. Specific actions include:

- **Rationalizing diagnostics:** Reducing unnecessary lab tests and imaging, which consume energy, materials, and chemicals. For instance, the UK's Academy of Medical Royal Colleges' campaign "**Choosing Wisely**" encourages doctors and patients to have conversations about reducing unnecessary treatments.
- **Opting for low-carbon treatments:** For example, using metered-dose inhalers for asthma has a high carbon footprint (due to potent propellant gases). Where clinically appropriate, switching to dry powder inhalers can reduce carbon emissions by over 90%.

- **Promoting telehealth:** For appropriate follow-up appointments, telehealth avoids patient travel emissions and reduces the need for physical space.

5.3.2 Sustainable Supply Chains:

Greening the supply chain involves extending the green procurement policy to engage directly with pharmaceutical and med-tech companies. For instance, the sector could demand transparency on the carbon footprint of products and reward suppliers with circular economy models (take-back, refurbishment, recycling).

5.3.3 Building Climate-Resilient Health Systems:

Building climate-resilient health systems prepares healthcare infrastructure for the impact of climate change (e.g., extreme heat, flooding, disease vector spread). This includes investments in:

- **Fortifying facilities** against extreme weather.
- **Diversifying energy sources** with backup renewables to ensure power during climate-induced grid failures.

6 Financing the "greening" of the health sector

Implementing the three-pillar approach outlined above requires resources to undertake assessments and fund the needed interventions. However, only a small proportion of Ministries of Health in low- and lower-middle-income countries (28%) are currently receiving international funds to support climate change and health work. This may explain why only 27 per cent of such countries have conducted assessments of the climate resilience of their health care facilities (WHO, 2021).

Notwithstanding the financial constraints, investments that decouple health expenditure from GHG emissions and align health care growth with the global decarbonization agenda can attract public and private climate financing. The potential is even greater for Africa's health sector, which heavily relies on fossil fuels to power operations. Investing in the reduction of health sector emissions is therefore not just an environmental imperative but a financial and operational one; such investments often attract climate funds.

6.1 Climate funding mechanisms[5]

Climate funding mechanisms such as carbon credits, green bonds, adaptation grants and results-based climate finance (RBCF) offer untapped resources for reducing health sector emissions (i.e., mitigation), building climate-resilient health infrastructure (i.e., adaptation) and scaling-up health sector financing.

Table 2. Potential health sector funding mechanisms

Mechanism	Application in Health Systems	Example

[5] The use of climate finance instruments and funding streams might also require relatively large-scale projects to ensure profitability. This can conflict with health goals in some cases where small, context-adapted interventions are most appropriate (Freeman and Zerriffi, 2012).

Carbon Credits	Renewable energy in hospitals, waste management	Solar-powered clinics earning offsets
Green Bonds	Financing eco-friendly health infrastructure	Ghana's health sector green bond
Adaptation Grants	Climate-proofing health facilities	Flood-resistant clinics in Mozambique
Results-Based Climate Finance (RBCF)	Pay-for-performance for emission reductions	Verified energy savings in South African hospitals

6.1.1 Carbon Offsets, Carbon Credits and Carbon Markets

Health sector investments or projects that verifiably reduce, avoid, or sequester an equivalent amount of carbon dioxide (CO2) or other greenhouse gases (GHGs) (i.e., carbon offsets) are eligible for **carbon credits** which can be sold for financial gain in **carbon markets.** The proceeds can be used to defray the upfront costs of green investments or to fund new health sector investments. In effect, carbon offsets[6] are **projects** that generate **carbon credits**[7] which can be **traded** for cash **in carbon markets**[8].

For instance, if a hospital in a low-income country installs a solar microgrid which reduces emissions due to lower diesel consumption, the verified emission reductions could then be sold as carbon credits, providing an additional revenue stream that pays back the initial investment faster.

Box 1: A Practical Pathway to implementing a carbon offset project

To scale investments using this model, a health organization should:

1. **Identify the Opportunity:** Conduct an energy audit or waste audit across multiple facilities.
2. **Secure Upfront Capital:** Partner with a carbon project developer, impact investor, or use a loan. The developer often provides the capital in exchange for a share of future credit revenue.
3. **Implement the Project:** Install the technology (solar panels, efficient lights, autoclaves, stoves).
4. **Measurement & Verification:** Work with a standard (like **Gold Standard** or **Verra**) to document the baseline, monitor the project, and have it third-party verified.
5. **Sell Credits:** The verified credits are sold on the carbon market to corporations or governments wanting to offset their emissions.
6. **Reinvest Revenue:** The carbon income is reinvested into core health services—paying staff, buying medicines, funding outreach programs, or financing the next round of upgrades.

[6] **Carbon offsets** are projects that reduce, avoid, or sequester an equivalent amount of carbon dioxide (CO2) or other greenhouse gases (GHGs) elsewhere.

[7] There is controversy regarding the effectiveness of carbon credits for climate change mitigation, their volatility and lack of transparency, as well as the power imbalances and processes of dispossession that can be driven by carbon projects and markets (Dunlap and Sullivan, 2020; Fleischman et al., 2021; Anguelovski and Corbera, 2023).

[8] Carbon markets are systems that create financial incentives to reduce greenhouse gas (GHG) emissions by **enabling the trading of carbon credits** or allowances. There are two main types: compliance markets, which are regulated and used to meet government-mandated emissions targets, and voluntary markets, where companies voluntarily purchase carbon credits to offset their emissions and meet net-zero goals.

6.1.1.1 Opportunities

Carbon credits are **not** typically a one-time payment upon project completion. Instead, they generate a **recurring revenue stream** over the lifetime of the project, which can be 10, 20, or even 30 years. This is possible because a carbon credit project operates on a cycle of **monitoring, reporting, and verification** (MRV). This cycle comprises the following three phases:

1. **Project Development and Registration (One-Time):** This is the initial, upfront phase where the project is designed, validated against a standard (like Verra or Gold Standard), and registered. This is costly and has no revenue.
2. **Crediting Period:** Once registered, the project enters a "crediting period," which is a fixed term (e.g., 7-10 years, often renewable) during which it is eligible to generate credits.
3. **The Revenue-Generating Cycle (Repeated):** Within this crediting period, the project operates on a continuous cycle of revenue generation.

6.1.1.2 Challenges

To qualify for carbon credits, carbon offset projects must demonstrate that they would not have been undertaken without the incentive of selling carbon credits (i.e., additionality). For instance, an energy efficiency upgrade project will only qualify for carbon credits if it can prove that the upgrade was too expensive or too risky to undertake, based on energy savings alone.

Secondly, the profitability of carbon offset projects depends on the prevailing price of carbon on international markets. Carbon offset projects in developing countries tend to attract relatively lower prices than in developed countries. Furthermore, volatility in carbon markets can also be an important drawback when it comes to securing long-term financing for health interventions (Hodge and Clasen, 2016). Carbon prices are determined by demand and supply conditions as well as the perceived credibility of the carbon offset project. Credibility can be enhanced by establishing transparent carbon registries and engaging credible entities to certify that the project has indeed achieved its objective. This leads to the third point.

The project must demonstrate that it achieved its objective of reducing carbon emissions. This is accomplished by engaging the services of a certification agency. Reputable certification entities include Verra and the Gold Standard[9].

6.1.1.3 Typical carbon offset projects

Investments in **renewable energy** (e.g., solar panels) **energy efficiency** and **waste management** can generate carbon credits and at the same time contribute significantly to the decarbonization of Africa's health sector. Such projects generate carbon credits because of their capacity to reduce greenhouse gas emissions.

[9] The Gold Standard for Global Goals has methodologies for verifying the authenticity of Energy Efficiency and Distributed Renewable Energy (DRE) projects.

Table 3: Health sector projects suited for carbon credit financing

Typology	Health Sector Example	Primary Health Co-Benefit
Energy Efficiency	Retrofitting a hospital with LED lights & efficient HVAC	Lower operational costs for healthcare; cleaner outdoor air
Waste Management	Replacing a medical waste incinerator with an autoclave	Prevents exposure to toxic emissions and infections
Renewable Energy	Solar panels & batteries for an off-grid health clinic	Reliable power for vaccines, lighting, and medical devices
Cross-Cutting	Distributing clean cookstoves to communities	Massive reduction in respiratory illness from household air pollution

6.1.1.3.1 Renewable Energy investments for Health Facilities

Replacing diesel generators with **solar mini-grids** is an example of an investment that is eligible for carbon credits under the Clean Development Mechanism**.** For instance, Rwanda's **Health Solar Initiative** reduced carbon emissions in health centers by 30 per cent using a blend of concessional carbon financing and carbon credits. The initiative used a **Global Climate Fund grant** (public) to cover 60% of costs, the remainder was private-sector financed using **carbon credit pre-purchases**.

Kaiser Permanente, a massive integrated health system in the USA, installed solar panels on hospital rooftops and carports and signed long-term power purchase agreements (PPAs) for utility-scale wind and solar farms. This locked in energy prices for decades, protecting them from volatile fossil fuel markets. Even though Kaiser Permanente **retired** the renewable energy credits (RECs) they generated to claim carbon neutrality (rather than selling them), this model shows that large-scale decarbonization is operationally and financially feasible.

6.1.1.3.2 Energy efficiency investments

Investments in energy efficiency can attract carbon credits as well as **adaptation funding**. **Energy efficiency can be achieved by** implementing Green Building Standards that mandate investments in retrofitting hospitals with energy-efficient designs (Leadership in Energy and Environmental Design (LEED) certification).

6.1.1.3.3 NHS Net Zero Strategy (United Kingdom)

The United Kingdom's National Health System (NHS), the world's first health system to commit to net-zero emissions (by 2040 for emissions it controls directly, and by 2045 for its wider supply chain) generated carbon credits through massive investments in energy efficiency. The strategy includes a £50 million investment to replace outdated hospital boilers with heat pumps and other low-carbon solutions, and a £330 million program to install energy-efficiency measures like LED lighting and improved building insulation across NHS sites. These investments are expected to yield £1.4 billion in energy savings by 2028 which can be reinvested into patient care.

6.1.1.3.4 Sustainable Medical Waste Management

The incineration of medical waste and the emission of anesthetic gases by health facilities are significant sources of greenhouse gas emissions. One ton of anesthetic gases, such as desflurane sevoflurane, is equivalent to 2,500 tons of CO_2. Medical waste is also energy-intensive to treat and often incinerated,

producing emissions and toxic ash. Investments that capture anesthetic gases and or lead to a shift from incineration of medical waste to **autoclaving/recycling** can generate carbon credits.

6.1.1.3.5 Autoclaving and microwave technologies

Many hospitals are investing in **autoclaving** (steam sterilization) and microwave technologies as a low-energy alternative to incineration for a large portion of infectious waste. Autoclaving is also significantly more energy-efficient compared to incineration. By using steam rather than combustion, autoclaves reduce the overall carbon footprint of medical waste disposal, which is especially important in an era where environmental sustainability is a priority. Moreover, autoclaving reduces waste volume by up to 80%, allowing for less material to be sent to landfills or incinerators, further contributing to environmental conservation (https://athermal.co.za/2025/04/15/autoclaving-in-medical-waste-management/).

Other medical waste management initiatives include rigorous waste segregation programs to reduce the volume of waste that requires high-energy treatment. This reduces emissions from incineration and can significantly lower waste management costs and reduce environmental pollution.

6.1.1.3.6 Capturing and Destroying Waste Anesthetic Gases

Anesthetic gases are potent greenhouse gases. A number of companies (e.g., Capture and SageTech) have developed technology to capture these gases from hospital vacuum systems. The captured gases are purified and can be recycled for future use or destroyed in a controlled manner, preventing their release into the atmosphere.

Carbon capture technology can be financed through a service model where the company installs the capture hardware, and the hospital pays a fee based on the volume of gas captured or the resulting carbon credits generated. This creates a direct link between a green investment and a reduction in the health sector's carbon footprint.

6.1.2 Green bonds

Green bonds are debt instruments that mobilize resources to fund or re-finance new or existing eligible **green projects**. They are typically issued by governments, development banks, or corporations. The resources mobilized through green bond issuances can be directed to health projects that have clear environmental benefits.

6.1.2.1 Opportunities:

Green bonds offer competitively priced climate financing at scale, because they facilitate access to large capital pools by tapping into the vast global demand for ESG (Environmental, Social, Governance) investments from institutional investors. Furthermore, due to high investor demand, green bonds can access capital at lower interest rates (i.e., a “green premium”).

6.1.2.2 Challenges:

The key challenges to the use of green bonds are eligibility and certification, a strict reporting burden and high upfront costs. Mobilizing climate financing through green bonds requires strict adherence to international standards (e.g., ICMA Green Bond Principles) to avoid "greenwashing." Projects must be verified as truly "green." Secondly, issuers must commit to transparent ongoing reporting on the use of proceeds and the environmental impact of the funded projects. Thirdly, the costs of certification and verification can be high. Notwithstanding these challenges, successful issuers of green bonds can deploy such resources to derisk and crowd-in private investors to the health sector.

6.1.2.3 Typical green bond projects include:

- Building new **energy-efficient, low-carbon hospitals and clinics**.
- Retrofitting existing health facilities with solar power, rainwater harvesting, and efficient HVAC systems.
- Electric or low-emission vehicle fleets for ambulances and health outreach.
- Sustainable medical waste management systems (e.g., non-burn autoclaves).

6.1.2.4 Green Bond Examples:

Nigeria was the first African country to issue a sovereign green bond, raising **₦10.69 billion** (about $30 million at the time) in 2017. While focused on agriculture, energy, and water, the framework explicitly included **off-grid solar electrification of rural clinics and hospitals** as an eligible category. This set a direct precedent for future issuances to directly fund climate-resilient health infrastructure. Egypt's **Leadership in Energy and Environmental Design (LEED)-Certified Hospitals** blended **green bonds** (private capital) with **government guarantees** (public capital) to reduce interest rates. These hospitals which are noted for their energy efficiency, water conservation, and use of sustainable materials include the Egyptian American Medical Center and the Children's Hospital.

6.1.3 Adaptation Funds (e.g., UN Adaptation Fund, Green Climate Fund)

Adaptation Funds are grants or highly concessional loans from dedicated climate funds to help developing countries adapt to the adverse effects of climate change. Established by the 16th Conference of Parties (COP16) in Cancun, Mexico, the Green Climate Fund (GCF) is the world's largest dedicated climate fund helping developing countries to reduce their greenhouse gas emissions and to enhance their ability to respond to climate change, in line with the Paris Agreement. The Fund is mandated to invest 50% of its resources in mitigation and 50% in adaptation in grant equivalent. At least half of its adaptation resources must be invested in the most climate vulnerable countries (Small Island Developing States (SIDS), Least Developed Countries (LDCs), and African States).

6.1.3.1 Financing instruments

GCF can structure its financial support through a flexible combination of grant, concessional debt, guarantees or equity instruments to leverage blended finance and crowd-in private investment for climate action in developing countries.

6.1.3.2 Opportunities:

Since adaptation funds are grants or highly concessionary loans, they have limited impact on the debt vulnerabilities of beneficiary countries even as they contribute to improving healthcare outcomes. Given their concessional nature, such funds are ideal for blending to derisk private investments. Adaptation funds often come with capacity building and technical assistance.

6.1.3.3 Challenges:

Accessing adaptation funds requires a highly competitive, complex and lengthy application process, especially since such resources are limited in quantity**.** Access to adaptation funds is further constrained by the limited number of accredited health institutions which can apply for climate funding, and barriers for small domestic applicants to compete with international accredited institutions (Borghi J, et., al, 2024).

Furthermore, a lack of awareness of the health climate nexus by multilateral climate fund managers has resulted in limited climate fund allocations to the health sector and vice versa. Only around 6% of adaptation funding and 0.5% of multilateral climate funding is currently allocated to projects that protect or improve human health (WHO 2023, Watts, Nick et al. 2018; WHO, 2018).

A lack of engagement between health and climate stakeholders and silos between respective Ministries also mean that countries often fail to include health within their National Adaptation Plans and Nationally Determined Contributions, which are pledges made by countries in the context of the Paris Agreement, detailing what they will do to help keep global heating under 1.5 degrees Celsius and adapt to the impacts of climate change (UNDP, 2023).

Implementation of adaptation projects requires strong national institutions to manage and report on the funds, which can be a challenge in some countries. Adaptation funds are project based hence cannot typically be used for recurrent costs like health worker salaries.

6.1.3.4 Typical adaptation fund projects in the health sector

Adaptation funds are **project-based** and ideally suited for projects where the **primary goal is health adaptation**, such as:

- **Early Warning Systems** for climate-sensitive disease outbreaks (e.g., malaria, cholera).
- **Strengthening surveillance and response** capacity for zoonotic diseases (spillover risk increased by climate change).
- **Climate-proofing health infrastructure:** Protecting clinics from floods, cyclones, or extreme heat.
- **Ensuring water security** for health facilities in drought-prone regions.

6.1.3.5 Senegal's Adaptation Fund Project

Notwithstanding the challenges in accessing adaptation funds, Senegal's Adaptation Fund Project: "Fighting Malaria and Climate Change in Senegal" represents a success story. The project was implemented by the Centre de Suivi Ecologique (CSE), which is accredited as a National Implementing Entity (NIE) for

the Adaptation Fund. The project aims to integrate climate information into the national malaria control program. This includes developing forecasts to predict malaria outbreaks based on rainfall and temperature patterns, allowing for proactive distribution of bed nets and medicines. This is a quintessential example of using adaptation finance for health system strengthening.

6.1.4 Results-based climate finance (RBCF)

Results-based climate finance is a conditional climate financing mechanism where payments are conditional on the achievement and verification of **pre-agreed results**. The most common type is carbon credit payments, however, RBCF's can also be used for verified adaptation outcomes. For instance, a funder (e.g., a climate fund) could contract a government to achieve a specific, verifiable health-adaptation outcome, such as reducing cholera incidence in a flood-prone region by a certain percentage. Payment is made upon verification of the desired outcomes.

Typical RBCF funded projects in the health sector include efficient cookstoves, renewable energy for clinics, and energy saving waste management practices.

6.1.4.1 Opportunities:

RBCBs incentivize outcomes by shifting the focus from inputs (spending money) to measurable results (lives saved, emissions reduced, diseases prevented). They also unlock innovation by giving implementers flexibility in how they achieve the agreed-upon result. RBCF's are a sustainable revenue model because they **c**reate a long-term income stream for health activities that have climate benefits (e.g., ongoing cookstove distribution).

6.1.4.2 Challenges:

RBCF's face a number of challenges including the complexity of verifying the outcomes of the projects, upfront financing costs and performance risk. Verification of the authenticity of green investments can be complex**,** requiring robust, independent systems to measure and verify the results, which can be expensive and technically demanding.

Implementers (e.g., a Health Ministry) must finance the activity upfront and only get reimbursed later, requiring access to bridge financing. Furthermore, implementers face p**erformance risks:** If results are not achieved (e.g., a drought affects the project), no payment is received, jeopardizing the investment.

6.1.4.3 BURN Manufacturing (Kenya) (Carbon Credit Model):

BURN Manufacturing (Kenya) is a private company in Kenya that manufactures and distributes clean cookstoves across Africa. Its projects generate millions of carbon credits from reducing deforestation and emissions. The revenue from these credits allows it to subsidize the cost of stoves for the poorest households, directly improving health outcomes (reduced pneumonia, etc.) and scaling its impact. A health ministry could replicate this model.

Table 4: Opportunities & Challenges

Instrument	Best For	Key Opportunity	Key Challenge	African Example
Carbon Credits	Renewable energy, energy efficiency, waste management.	Creates a sustainable revenue stream; directly monetizes the climate benefits of health	Complex, expensive validation & monitoring; requires long-term verification;	A Ugandan health NGO funds its clean cookstove distribution

		interventions; strong potential for scale.	reputational risk around credit quality ("greenwashing").	program via revenue from carbon credits sold on the voluntary market.
Green Bonds	**Mitigation-focused infrastructure:** Building/retrofitting green, energy-efficient health facilities.	Access to large-scale capital from private investors.	Complex certification process to avoid "greenwashing".	Nigeria's Sovereign Green Bond (eligible category).
Adaptation Funds	**Adaptation-focused programs:** Early warning systems for diseases, climate-proofing clinics.	Grant-based financing, no debt.	Highly competitive and complex application process.	Senegal's climate-informed malaria control program.
RBCF	**Interventions with measurable outcomes:** Cookstove distribution, off-grid solar, disease reduction.	Incentivizes efficiency and proven results; creates sustainable revenue.	Complexity of verifying the outcomes of the projects, upfront financing costs and performance risk	BURN Manufacturing (Kenya) generates millions of carbon credits from reducing deforestation and emissions

6.2 Financing transformative green health sector investments[10]

Section 5 identified investments in energy efficiency, renewable energy and waste segregation and reduction as critical areas where green investments can have a transformational impact on the carbon footprint of the health care sector. Section 6 highlighted key climate funding mechanisms that can be directed to financing green investments in the healthcare sector.

Building on the above discussions, this section identifies appropriate instruments to finance green healthcare interventions in the critical areas of renewable energy; energy efficiency; sustainable waste management; climate resilient health infrastructure; low emissions vaccine cold chains, and green hospitals construction/retrofits. The climate financing mechanisms are carbon credits, green bonds, PPPs and concessional financing instruments such as climate adaptation funds.

The full potential of these disparate funding streams is realized when they are blended to derisk and crowd in private financing. Resources from government-backed green bond issuances can be blended with private funds to finance large scale solar projects. Concessional financing from sources such as the GCF can help finance the high upfront costs associated with carbon offset projects initiated by the private sector. Collectively, such initiatives can forge enduring Public Private Partnerships with the co-benefits of reducing the operational costs and carbon footprint of health facilities while creating jobs and revenue streams for investors.

[10] https://www.greenclimate.fund/about. GCF operates through a network of over 200 Accredited Entities and delivery partners who work directly with developing countries for project design and implementation.

Table : Intervention-Funding Alignment

Intervention	Best Funding Mechanism	Private Interest?	Blended Finance Viable?
Renewable energy - Solar health clinics	Carbon credits + concessional loans	Yes (high)	Yes
Medical waste recycling	RBCF + PPPs	Yes (moderate)	Yes
Flood-resistant clinics	Adaptation grants	No	Limited
Vaccine cold chains	Carbon credits + aid	Yes (high)	Yes
Green hospitals	Green bonds + PPPs	Yes (high)	Yes

6.2.1 Financing mechanisms for Green Health Projects

6.2.1.1 Renewable Energy Projects

Blending carbon credits and concessional loans is an effective financing mechanism for renewable energy projects in the health sector. Concessional climate financing can be used to finance upfront costs of the project pending the realization of the future income stream generated by the carbon credits over the life of the project.

Table 5: Funding Mechanisms: Renewable energy and energy efficient hospitals/clinics

Mechanism	Applicability	Public/Private Interest	Blended Potential?
Carbon Credits	High (verified emission reductions from diesel displacement).	**Private** (corporates, carbon brokers).	Yes (combine with public grants for upfront costs).
Green Bonds	Viable for large-scale projects (e.g., national hospital solarization programs).	**Mixed** (sovereign bonds attract private investors).	Yes (e.g., IFC-backed bonds).
Concessional Loans	Critical for high upfront costs (e.g., AfDB loans at low interest).	**Public** (MDBs, governments).	Yes (blend with grants).
PPP Models	Private firms design/build hospitals in exchange for long-term leases.	**Private** (construction companies).	Yes (public land contributions).

Green bonds are suited to financing large scale renewable energy projects such as national hospital solarization programs. The financing generated through green bond issuances can be leveraged by a PPP arrangement. **Egypt's LEED-Certified Hospitals** blended **green bonds** (private capital) with **government guarantees** to reduce interest rates. These Egyptian hospitals include Magdi Yacoub Global Heart Centre which has achieved LEED Gold certification while the new National Cancer Institute (NCI), is targeting LEED Gold certification.

6.2.1.2 Medical waste recycling

Results Based Climate Funds and PPPs have proven effective for financing medical waste recycling projects. **Ghana's Korle-Bu Hospital** partnered with **Zoomlion** (a private waste firm) under a PPP, with partial **government subsidies** for low-income regions.

Table 6: Funding Mechanisms for waste management (Shift from incineration to autoclaving/recycling)

Mechanism	Applicability	Public/Private Interest	Blended Potential?

Results-Based Climate Finance (RBCF)	Payments for verified waste emission reductions (e.g., methane capture).	**Mixed** (public funders + private verifiers).	Yes (e.g., World Bank's Carbon Initiative for Development).
PPP Models	Private operators manage waste for fee-for-service (e.g., South Africa's BioTherm).	**Private** (waste management firms).	Yes (public subsidies for rural areas).
Adaptation Grants	If linked to disaster resilience (e.g., flood-safe waste storage).	**Public** (GCF, Adaptation Fund).	Limited.

6.2.1.3 Climate resilient health infrastructure

Flood resistant clinics can be funded through a combination of adaptation grants, catastrophe bonds and domestic health budgets. Catastrophe bonds are high-yield insurance-linked securities that transfer a specific, defined catastrophic risk from a sponsor to investors. In exchange for a high-interest coupon, the investors agree to lose all or part of their principal if a major disaster, such as a hurricane or earthquake, occurs. **Mozambique's Cyclone-Resilient Clinics** are funded by **Adaptation Fund grants** (100% public), but future projects could blend with **insurance-linked securities.**

Funding Mechanisms: Climate resilient health infrastructure

Mechanism	Applicability	Public/Private Interest	Blended Potential?
Adaptation Grants	Primary source (e.g., GCF, Adaptation Fund).	**Public** (grant-dependent).	Limited (rarely blended).
Catastrophe Bonds	For disaster-prone regions (e.g., African Risk Capacity).	**Private** (investors seeking high yields).	Yes (public risk guarantees).
Domestic Health Budgets	Co-financing for locally led projects.	**Public** (limited private role).	No.

6.2.1.4 Low-Emissions Vaccine Cold Chains

Carbon credits coupled with concessional financing, including ODA, are effective financing sources for vaccine cold chains. The PPP model involves a private company financing, installing, and maintaining the energy system, while the public sector pays for the *service*.

A service provider (often a specialized solar company) owns, operates, and maintains the solar power system and the cold chain equipment on-site at a health clinic. The government or health agency then pays a regular, predictable fee for the *cooling service* or the *energy output* (kWh), not the physical assets.

In addition to the service fees earned by the investor, additional revenues accrue from carbon credits earned by reducing or avoiding emissions using solar or other green energies to power the cold chain project. Carbon credits create a new revenue stream that helps offset the high upfront costs of the renewable energy project as well as fund operational costs.

Funding Mechanisms: Solar-powered refrigeration for vaccine storage.

Mechanism	Applicability	Public/Private Interest	Blended Potential?
Carbon Credits	High (verified cuts from diesel-powered cold chains).	**Private** (pharma CSR, offset buyers).	Yes (Gavi's cold chain optimization fund).
Development Aid/Concessionary financing	Key for low-income countries (e.g., Gavi, Global Fund).	**Public** (donor-driven).	Yes (match with private logistics firms).

PPP's	Private operators manage for fee-for-service.	**Private** .	Yes (public subsidies for rural areas).

This innovative financing model removes the massive technical and maintenance burden from health ministries and completely eliminates upfront capital costs for the government. The private company's profit is tied to the system's reliability and uptime, creating a strong incentive for quality. This in turn provides long-term operational certainty and ensures professional maintenance, which is a common failure point for solar systems in remote areas. Companies like **Engie Energy Access** and **M-KOPA** are already providing solar home systems on a pay-as-you-go model. **Nigeria's Solar Direct-Drive Fridges** (with **Gavi** and **UNDP**) combined **grant funding** with **carbon revenue** from diesel displacement. A similar model can be applied to larger-scale solar solutions for health clinics.

7 Conclusions

Africa's health sector is underfunded, has suboptimal outcomes and is a significant source of greenhouse gas emissions due in part to high dependence on diesel generators. However, financing initiatives that green the health sector is highly viable and increasingly urgent. The key is to view "greening" not as a cost, but as a strategic investment in a health system's financial and operational sustainability. The most successful examples, like Kaiser Permanente and the NHS, show that the business case is built on long-term operational savings and risk mitigation. By viewing health infrastructure through the lens of carbon mitigation, health agencies can turn operational costs (like energy bills and waste disposal) into revenue-generating assets that fund better patient care and public health programs.

Carbon credits, green bonds and concessionary financing can serve as useful accelerators for investments in renewable energy, energy efficiency and waste management. Concessionary green financing such as adaptation funds can be used to derisk and crowd in private investments in carbon offset projects that generate a revenue stream of carbon credits.

Private sector investor interest is strong for renewable energy, waste management, and green construction (profit potential via carbon credits/PPPs). Blended financing approaches work best for projects with revenue streams (e.g., carbon credits, energy savings) and high upfront costs (e.g., solar clinics) requiring public concessional capital to de-risk and crowd-in private investment.

Projects that are unattractive to the private sector (e.g., climate resilience, disaster-proofing) due to their low returns on investment are ideal for the public-sector and should rely on grants and concessionary financing.

8 Policy Recommendations:

To build a healthier, more resilient Africa in the face of climate change, a strategic and integrated approach is essential. This journey begins by strengthening the policy foundation. African governments can lead by weaving climate action directly into the fabric of their national health plans. This means

revising National Health Strategic Plans to explicitly include climate resilience and low-carbon strategies, ensuring they are fully aligned with the country's commitments under the Paris Agreement. To fuel this transition, establishing a dedicated Climate-Health Financing Facility under a continental body like the Africa CDC or the African Union would be a powerful step, creating a central hub to blend climate and health funding. A promising and innovative path forward is to pilot carbon credit projects, allowing hospitals to partner with voluntary carbon markets and generate revenue by reducing their own emissions.

Mobilizing finance is the next critical step. African governments can attract private capital by prioritizing health projects that generate carbon credits, while using adaptation grants for essential, non-revenue-generating resilience upgrades. Establishing public-private partnership (PPP) units can help broker these complex deals, following successful models like Rwanda's energy-performance contracts. For their part, development partners can amplify these efforts by supporting blended finance instruments that combine grants, concessional loans, and future carbon revenue. They can also provide vital technical assistance, particularly in building robust systems for measuring, reporting, and verifying (MRV) carbon credits in the health sector.

The true power of this strategy lies in seamlessly integrating climate and health financing. For African health ministries and donors alike, this process starts with data. Conducting thorough vulnerability and adaptation assessments will pinpoint the biggest climate-health risks and identify which funding streams are best suited to address them. Next, we must build integrative capacity by forming cross-Ministerial teams where health officials learn the language of climate finance and climate fund managers deepen their understanding of health outcomes. The most effective way to build momentum is to start with a tangible pilot project—such as solarizing a single clinic funded by a carbon project—to demonstrate proof-of-concept before scaling up. Throughout this process, leveraging partnerships with experienced carbon project developers, development banks, and NGOs will be crucial for navigating these complex transactions.

By strategically wielding these instruments, the African health sector can unlock billions of dollars in new funding. The immediate next steps to set this in motion are to conduct a feasibility study on the carbon credit potential within African hospitals and to convene a regional inter-Ministerial task force—bringing together leaders from Health, Environment, and Finance—to develop a concrete roadmap. This comprehensive approach promises to build healthcare systems that are not only healthier for people, but also more resilient in the face of the defining crisis of our time.

8.1.1 Typology of Innovative Climate Finance Instruments for Health System Strengthening

Instrument	Core Mechanism	How it Funds Health Strengthening	Key Opportunity	Key Challenge	Potential African Example
Carbon Credits	Payment for Verified Emissions Reductions: Sale of certified units (credits) each representing 1 tonne of CO_2e avoided or removed.	Generates revenue: Health projects (efficient cookstoves, solar clinics, waste management) generate saleable credits. Finances interventions: Revenue funds the health activity itself (e.g., stove distribution).	Creates a sustainable revenue stream; directly monetizes the climate benefits of health interventions; strong potential for scale.	Complex, expensive validation & monitoring; requires long-term verification; reputational risk around credit quality ("greenwashing").	A Ugandan health NGO funds its clean cookstove distribution program via revenue from carbon credits sold on the voluntary market.
Green Bonds	**Use-of-Proceeds Bond:** Raises debt for pre-defined eligible green projects.	Funds capital projects like building energy-efficient, solar-powered hospitals or retrofitting clinics with renewable energy and water harvesting.	Accesses large pools of private capital from ESG investors; can offer lower interest rates.	Strict certification required to avoid "greenwashing"; reporting burden on use of proceeds.	A Nigerian state issues a bond to solarize all 50 primary healthcare centers in its capital.
Adaptation Funds	**Project-Based Grants:** Provides grants for specific projects that build climate adaptation resilience.	Funds programs like early warning systems for climate-sensitive diseases (malaria, cholera) or "climate-proofing" clinics against floods/cyclones.	Grant-based (no repayment); specifically designed for adaptation needs; includes technical support.	Highly competitive and complex application process; not for recurrent costs.	Senegal's project to integrate climate data into its national malaria control program.
Results-Based Climate Finance (RBCF)	**Payment for Verified Results:** Pays out after pre-agreed, verified outcomes are achieved.	**Carbon Credits:** Revenue from health projects (efficient cookstoves, solar clinics). **Adaptation:** Payment for reducing cholera cases after flood defenses built.	Incentivizes efficiency and outcomes; creates a sustainable revenue stream for health-climate activities.	Requires robust verification; implementers need upfront capital; performance risk if results aren't met.	A health NGO gets paid per verified ton of CO2 avoided by its clean cookstove program in Malawi.
Sustainability-Linked Bonds (SLBs) & Loans	**Financial Incentive for Performance:** General-purpose debt where interest rate is tied to achieving Sustainability Performance Targets (SPTs).	A hospital network raises capital for general use. Its loan interest rate decreases if it hits SPTs like reducing GHG emissions by 25% or powering 50% of facilities with renewables.	Flexible use of funds; creates a direct financial incentive to achieve strategic	Sophisticated tracking of SPTs needed; financial penalty (higher interest) if targets are missed.	A pan-African hospital chain takes an SLB, tying its interest rate to a target of cutting

			climate-health goals.		emissions across all its facilities.
Debt-for-Climate/Sustainability Swaps	**Debt Restructuring for Investment:** A portion of external debt is forgiven in exchange for domestic investment in climate/environmental programs.	Freed-up fiscal resources are channeled into a national fund for climate-health priorities, e.g., a national solar clinic roll-out or climate-proofing key hospitals.	Converts debt burden into domestic investment in resilience; addresses fiscal space constraints.	Extremely complex negotiations; requires high-level political engagement and strong governance.	Cape Verde negotiates a swap, agreeing to invest the forgiven debt amount in storm-resistant health infrastructure.
Blended Finance Funds	**De-risking with Concessional Capital:** Uses public/donor funds to absorb risk and attract private investment to thematic areas.	A fund provides equity and debt to private companies providing climate-health solutions, e.g., solar fridge manufacturers or modular clinic builders.	Mobilizes large-scale private capital that would not otherwise invest in "risky" climate-health projects.	Complex to structure; requires aligning differing return expectations of public and private investors.	A DFI-funded "Climate-Health Resilience Fund" invests in a Kenyan startup producing off-grid solar solutions for clinics.
Climate Insurance & Risk Pooling	**Parametric Payouts:** Provides automatic, rapid payouts triggered by pre-defined climate events (e.g., rainfall, wind speed).	Provides immediate liquidity to a health ministry for emergency response after a cyclone or flood, before traditional aid arrives, ensuring continuity of care.	Predictable, rapid financing for disasters; protects health budgets; promotes fiscal resilience.	Designing accurate triggers is complex; premiums can be costly (though pooling helps).	A health ministry buys parametric cyclone insurance through a regional risk pool, ensuring funds for emergency response.
Development Impact Bonds (DIBs)	**Outcomes-Based Contracting:** Private investors front capital for interventions; outcome funder (donor/govt) repays them only if results are achieved.	Investors fund a project to install water purifiers to reduce diarrhea in flood-prone areas. A donor repays them only if disease rates fall by a verified amount.	Shifts performance risk from public sector to investors; focuses entirely on achieving results.	Most complex to structure; high transaction costs; dependent on measuring complex health outcomes.	A DIB in Mozambique funds a project to reduce malnutrition rates in drought-affected communities.

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